

Alyssa Gao

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EDUCATION

Barnard College, Columbia University | New York, NY

Expected May 2027

BA in Computer Science with minors in Statistics and Political Science

GPA: 3.7

Relevant Coursework: Advanced Programming (C), Data Structures (Java), Voting & American Politics, Usable Security & Privacy, Applied Bayesian Analysis, Pol Quant Research Design (R)

University of Illinois, Urbana-Champaign | Urbana, IL

Aug 2023-May 2024

Major in Political Science and Statistics

GPA: 3.9

Relevant Coursework: Statistical Analysis (R), Intro to Pol Research

HONORS & GRANTS

The Andrew W. Mellon Foundation Enhancing the Sciences Fund at Barnard College (2026)

CyberNYC Research Experiences for Undergraduates Funding (2026)

Dean's List, Barnard College (2026)

Dean's List, University of Illinois, Urbana-Champaign (2024)

RESEARCH EXPERIENCE

What To Do When Your Phone Gets Stolen:

A Systematic Evaluation of Online Advice for Phone-Theft Victims

Research Fellow | *USENIX Security 2027 (in prep)*

May 2026 - Present

Advisors: [Professor Lucy Simko](#), [Professor Yasemin Acar](#) (Paderborn University, Germany)

- Co-leading a research team evaluating the quality, coverage, and actionability of online advice available to phone-theft victims.
- Designed a search protocol to emulate how a victim would seek help, authoring 30 generic and 60 scenario-specific queries and controlling for time, location, and personalization to reduce confounds.
- Reached saturation on generic queries by running three complete search rounds and tracking source-growth decay; collected first-page results across all queries to build a corpus of ~300 sources.

- Applied reflexive thematic analysis to develop an inductive codebook with two schemes (post-theft recovery goals and pre-theft threats), then coded sources into a source × code matrix to expose systematic coverage gaps.
- Key findings: advice protects the device but not the person, instead it's strong on immediate device threats, but weak on financial, data, and emotional harm; emotional support is nearly absent from official pages and appears almost only in peer forums; and no single source is comprehensive.

Auto Cookie Rejector:

A Chrome Extension for Automated Consent Refusal

Course research project | JavaScript · Manifest V3 · Privacy UX

Jan 2026 – May 2026

Advisor: [Professor Lucy Simko](#)

- Built a Manifest V3 Chrome extension (service workers, content scripts) that detects cookie consent interfaces and automatically selects the reject option, extending prior qualitative work into a functional privacy tool.
- Engineered resilient DOM querying with fallback logic to handle dynamic loading, iframes, and site-specific consent implementations.
- Evaluated across 10+ high-traffic e-commerce and media sites; characterized failure modes concentrated in iframe-embedded and shadow-DOM banners, revealing which design patterns most resist automated refusal.
- Connected technical findings back to the dark-patterns research: interfaces hardest to automate away often correspond to the most manipulative consent designs.

Cookie Consent Interfaces:

A Qualitative Study of Dark Patterns and User Decision-Making

Course research project | Usable Privacy · HCI · Qualitative Methods

Sep 2025 – Dec 2025

Advisor: [Professor Lucy Simko](#)

- Investigated how interface design and dark patterns shape user privacy decisions on cookie consent banners, situating the work in the gap between GDPR's definition of valid consent and its real-world technical implementation.
- Designed and ran a qualitative study: conducted semi-structured interviews with 5 participants, developed an iterative codebook, and performed thematic analysis to surface patterns in how users interpret and respond to consent interfaces.
- Identified recurring manipulative design mechanisms (unequal button salience, forced scrolling, and pre-checked defaults) and characterized how each undermines the voluntariness that consent regulation presumes.
- Synthesized findings into a research poster and written report translating interface-level observations into design and policy implications for meaningful consent.

Pokharel, A.*, **Gao, A.***, Irungu, A., Acar, Y., & Simko, L. (2026). *What To Do When Your Phone Gets Stolen: A Systematic Evaluation of Online Advice for Phone-Theft Victims* [Poster]. Barnard College Summer Research Institute, New York, NY. *Equal contribution.

Nasser, S., **Gao, A.**, & Waldron, J. (2025). *Bob Already Decided: Habit and Intrusion in Cookie Consent* [Poster] [Paper]. COMS BC3422, Barnard College, New York, NY.

TECHNICAL SKILLS

Computational: R, Python, Java, JavaScript, C

LANGUAGES

English (fluent), Chinese-Mandarin (Native), Cantonese (Native)